

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Geology and Geophysics

2. Subject Name: Structural Geology L-T-P: 3-1-0 Credits: 4

3. Pre-requisites: *Introduction to Earth Sciences*

4. Syllabus and reference books: [U1]

Syllabus:

Concepts of Stress and Strain in Structural Geology (including elastic, plastic and viscous materials; parameters controlling ductility/strength of materials, Mohr Stress circle; strains associated with homogeneous deformation, basics of plane strains and its graphical representation, shear strain calculation, problem solving); Folds (basic concepts, classification, mechanisms; interpretation of geological maps with folds); Foliation (primary vs secondary foliation, types of foliations, processes of foliation development, crenulation cleavage development, importance of foliation/cleavage and bedding plane relationship in structural geology, relation between foliation and principal axes of strain ellipsoid); Lineations (classification, linear structures, boudin formation, strain ellipsoid and stretching lineation); Faults (classification, normal/strike-slip/reverse faults and thrusts, relation between principle stress axes orientation and faulting, thin skinned vs. thick skinned tectonics, criteria to identify faults in the field, different fault rocks and deformation conditions for their formation, interpretation of geological maps with faults). Joints (classification, importance of ornamentations associated with extensional joints, conjugate joints and stress orientations); Unconformities and basement cover relationship (including examples from Precambrian rocks of India, interpretation of geological maps with unconformity); introduction to shear zones and superposed deformation; structures associated with igneous rocks, diapirs and salt domes; introduction to porphyroblast-matrix relationship in structural geology; Equal Area Projections and Orthographic Projections (tutorial exercises).

Reference Books:

- 1) Structural Geology (Haakon Fossen, 2010).
- 2) Structural Geology (Twiss and Moores, 2007).
- 3) Structural Geology of Rocks and Regions (Davis and Reynolds, 1996).
- 4) An Outline of Structural Geology (Hobbs, Means and Williams, 1976).
- 5) Structural Geology (Ghosh, 1993).
- 6) Microtectonics (Passchier and Trouw, 2005).
- 7). Basic Methods of Structural Geology (Marshak and Mitra, 1988)
- 8). Structural Geology: An Introduction to Geometrical Techniques (Ragan, 2009).

5. Lecture-wise break-up: [U2][U3]

Sl. No.	Topic	No. of lectures
1.	Concepts of Stress and Strain in Structural Geology (with tutorials/exercises)	10
2.	Folds (with geological maps)	8
3.	Foliation (with equal area projection exercises)	5
4.	Lineations (with equal area projection)	4

	exercises)	
5.	Faults (with geological maps)	7
6.	Joints	3
7.	Unconformity & basement cover relation (with geological maps)	3
8.	Introduction to Shear Zones and Superposed Deformation	3
9.	Structures associated with igneous rocks	1
10.	Introduction to porphyroblast-matrix relationship in structural geology	1
11.	Orthographic projections (exercises)	2
Total number of hours (<i>including tutorials</i>)		47